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Module 1 Test questions and your answers

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1. For each of the wavelength ranges **8 / 8 points**

1. 0.4-2.5μm
2. 3-5μm
3. 10-12μm

choose the correct statements below

- [x] 0.4-2.5μm is good for assessing vegetation condition

  Correct
  Correct. Vegetation condition is most easily assessed in the visible and near infrared ranges. Not much can be told about vegetation at the other wavelengths.

- [ ] 10-14μm is good for monitoring forest fires
- [x] 10-14μm is good for detecting warm objects at night

  Correct
  Correct. Objects around room temperature, including the earth itself, emit well in this range of wavelengths.

- [x] 3-5μm is good for monitoring forest fires

  Correct
  Correct. These wavelengths are best for detecting the strong emission from very hot objects, such as fires.

- [ ] 3-5μm is good for assessing water quality
- [x] 0.4-2.5μm is good for assessing water quality

  Correct
  Correct. Water condition can only assessed in the visible and near infrared ranges. Water is strongly absorbing at the other wavelengths, as we will see later in the course.

2. A multispectral scanner has been designed for aircraft operation. It has an FOV of +/-35 degrees about nadir, and an IFOV of 2mrad. The sensor has been designed for a flying height of 1000m. Note that an angle of A radians projects a line of length L=hA metres, at a distance of h metres. **6 / 6 points**

Choose all the correct answers.

- [ ] pixel size is 1m
- [x] pixel size is 2m

  Correct
  Correct. Pixel size=IFOV x flying height = 2x10-3 x 1000 = 2m

- [x] number of pixels across a scan line is 611

  Correct
  Correct. The most reliable way of calculating this is to divide the FOV (70 degrees) by the IFOV (in consistent units). Thus number of pixels per scan line = 70 x pi/180 / 2x10-3 = 611.

- [ ] number of pixels across a scan line is 620
- [x] swath width is 1222m

  Correct
  Correct Swath width=FOV x flying height = 70 x p/180 x 1000 = 1222.

- [ ] swath width is 1242m
- [ ] swath width is 2222m

3. Which of the following answers are true? **4.800000000000001 / 6 points**

- [ ] Earth curvature leads to significant geometric distortion for satellites with a small field of view.
- [x] Earth curvature leads to significant geometric distortion for satellites with a wide field of view.

  Correct
  That is correct. Earth curvature is a problem for a large FOV sensor.

- [x] Platform attitude variations are significant sources of distortion for aircraft based remote sensing imaging.

  Correct
  That is correct. Aircraft attitude can be significantly affected by wind and atmospheric turbulence, leading to geometric errors

- [ ] Earth rotation is not a problem for satellite imagers with a wide field of view.
- [ ] In general, earth rotation is not a problem for aircraft imaging.

  You didn't select all the correct answers

4. Suppose a particular sensor was capable of imaging in the following wavebands: **10.5 / 12 points**

1. 0.45-0.52μm (blue)
2. 0.52-0.60μm (green)
3. 0.63-0.69μm (red)
4. 0.76-0.90μm (near IR)
5. 1.40-1.45μm (mid IR)

Choose all the correct answers.

- [ ] band 1 is good for noting the loss of chlorophyll resulting from dying vegetation
- [x] band 3 is good for noting the loss of chlorophyll resulting from dying vegetation

  Correct
  That is correct. Band 3 (red) is very sensitive to loss of chlorophyll

- [ ] band 3 is best for discriminating between vegetation and water
- [x] band 4 is best for discriminating between vegetation and water

  Correct
  That is correct. In the near IR band water is almost black whereas vegetation is generally very bright

- [x] band 5 is best for monitoring the loss of moisture from soil

  Correct
  That is correct. The mid-IR range is very sensitive to moisture content

- [ ] band 1 is best for monitoring the loss of moisture from soil
- [ ] band 4 is best for discriminating between soil and vegetation
- [ ] band 3 is best for discriminating between soil and vegetation

  You didn't select all the correct answers

5. The spectral space in Figure 1 of the information sheet contains pixels of vegetation, water and non-vegetated regions (largely cleared land, roadways, etc.). Choose all the correct answers. **6 / 6 points**

- [ ] spot A corresponds to vegetation
- [x] spot B corresponds to vegetation

  Correct
  That is correct. Vegetation is bright in the NIR and dark in the visible red

- [ ] span C corresponds to vegetation
- [x] spot A corresponds to water

  Correct
  That is correct. Water is dark in both the NIR and red responses

- [ ] spot B corresponds to water
- [ ] span C corresponds to water
- [ ] spot A corresponds to non-vegtated covers other than water
- [ ] spot B corresponds to non-vegetated covers other than water
- [x] span C corresponds to non-vegetated covers other than water

  Correct
  That is correct. Non-vegetated cover types spread along the line which indicates similar NIR and red responses.

6. Equation 1 in the information sheet shows the covariance matrices for two images. Which image would show the most change from a principal components transformation? Choose one answer. **4 / 4 points**

- (●) image 1
- ( ) image 2

  Correct
  That is correct. The first covariance matrix corresponds to an image with a higher degree of correlation than the second one. Thus, a principal components transformation of the first image would show a greater change from the original than one computed on the second image.

7. Why do colour image products formed from principal components images generally appear richer in colour than a colour composite formed from the original bands of a remote sensing image? Choose the single correct answer from the following. **8 / 8 points**

- ( ) The high degree of correlation among the principal components means they can fill the colour space better.
- (●) The lack of of correlation among the principal components means they can fill the colour space better.

  Correct
  That is correct. When uncorrelated bands are contrast enhanced they can fill the colour space creating a better range of hues.